ASCEND No-GPS Drone Stack, Wiring, and 2-Day Build Guide

Final working architecture for Cube Orange + Raspberry Pi 5 + optical flow + rangefinder + camera + pogo-pin base station. Designed to be buildable in real life and easy to assemble under a 2-day schedule.

System	Final choice
Flight controller	Cube Orange with ArduCopter / Mission Planner
No-GPS stability	Optical flow + rangefinder + EKF2
Companion compute	Raspberry Pi 5
Detection	Pi Camera v3 or USB camera + OpenCV
Data transfer	Wi-Fi + rsync/scp over SSH
Base station power	24V 5A SMPS -> CC/CV charging stage -> pogo pins

The uploaded component sheet currently includes a GPS module, telemetry radio, and a heavy RPLIDAR A3M1. For the 2-day GNSS-denied build, GPS is removed, telemetry radio is optional, and heavy SLAM is skipped. The core build remains flight controller, Raspberry Pi 5, optical flow, rangefinder, camera, propulsion, battery, and docking/charging hardware.

1. End-to-end workflow

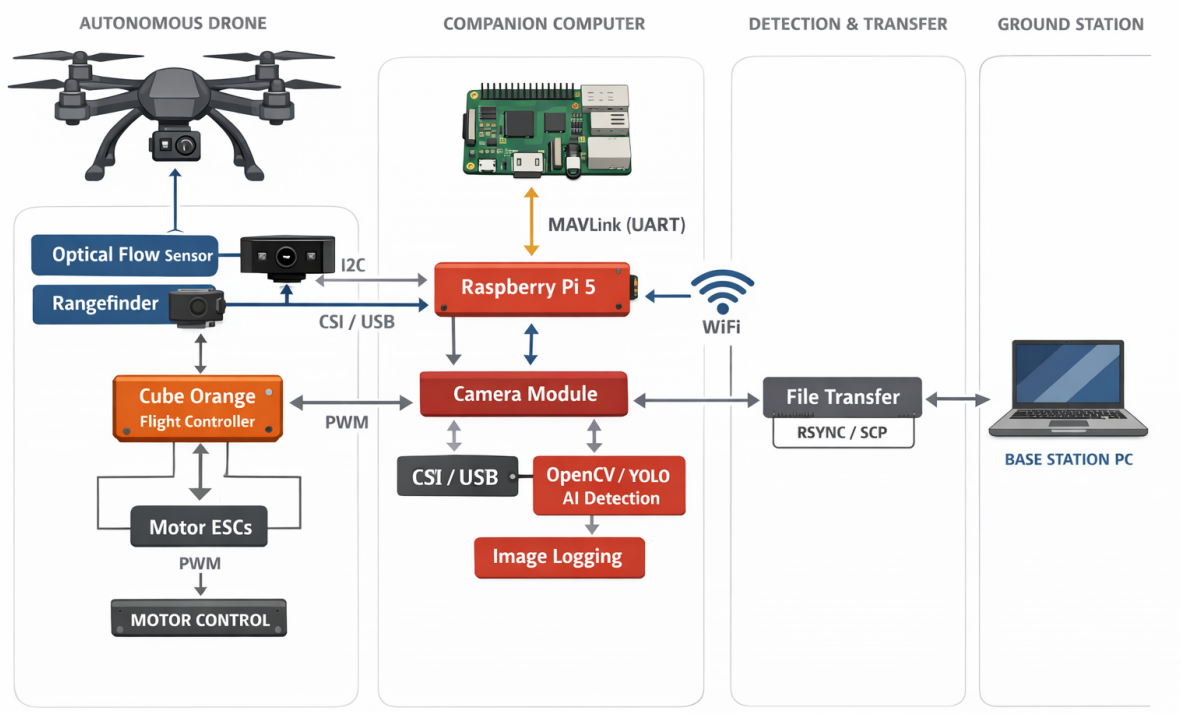

Figure 1. Flight control, companion computer, detection, and transfer flow.

2. Final connection port map

From	Interface / Port	To	Purpose	Build note
Cube Orange	TELEM1 (UART)	Raspberry Pi 5	MAVLink telemetry / command channel	Use a short shielded cable; keep baud rate fixed.
Cube Orange	PWM outputs	ESCs	Motor control	One output per motor.

From	Interface / Port	To	Purpose	Build note
Cube Orange	I2C	Optical flow sensor	X-Y velocity / indoor position hold	Mount rigidly and keep sensor facing ground.
Cube Orange	I2C or UART	Rangefinder / ToF	Altitude hold and landing	Place near center, pointed straight down.
Raspberry Pi 5	CSI	Pi Camera v3	Target detection frames	Use ribbon cable with strain relief.
Raspberry Pi 5	Wi-Fi / SSH	Base station laptop	Image and log transfer	Use rsync as primary transfer method.
Base station	24V DC input	CC/CV charger stage	Charging power source	Use 24V 5A SMPS for a 3A battery charge path.
Charger stage	Pogo pin contacts	Drone battery dock pads	Battery charging	2 power pins + 2 return pins minimum.
Base station controller	Relay / MOSFET enable	Charging rail	Charge interlock	Enable only after dock confirmation and disarm.
Base station fan	12V or 5V rail	Cooling fan	Dock cooling / battery bay airflow	Add filter mesh if dust is expected.

3. Software stack

Layer	Software / Library	Use
Flight stack	ArduCopter	Autopilot firmware on Cube Orange
Ground station	Mission Planner	Arming, calibration, tuning, failsafe setup
Companion runtime	Python 3	Main application runtime on Raspberry Pi 5
MAVLink	pymavlink	Vehicle state, battery, command and telemetry access
Vision	OpenCV	Frame handling, basic detection, image processing
Camera	picamera2 / libcamera	Camera capture on Raspberry Pi OS
Data	NumPy	Frame arrays and simple processing
Transfer	rsync / scp over SSH	Image and log movement to base station
Logging	logging / CSV / SQLite	Mission record and evidence capture

4. Base station build blocks

Module	Recommended implementation	Notes
Charging source	24V 5A SMPS	Enough input power for a 4S 3000 mAh charge path; use headroom.
Charging stage	CC/CV buck charger set to 16.8V	Charge current target 3A for 4S 3000 mAh.
Interlock	Relay or MOSFET enable	Power only after docking and disarm confirmation.
Contacts	Gold-plated pogo pins	At least 4 pins: 2 positive, 2 negative.
Alignment	Funnel / guide walls + magnets	Makes real-life docking easier and safer.
Compute	Base station laptop / PC	Stores logs, images, and reports.
Cooling	Fan + filter mesh	Useful for charger and compartment airflow.
Optional future	Solar panel + battery backup	Add later; not required for the first 2-day demo.

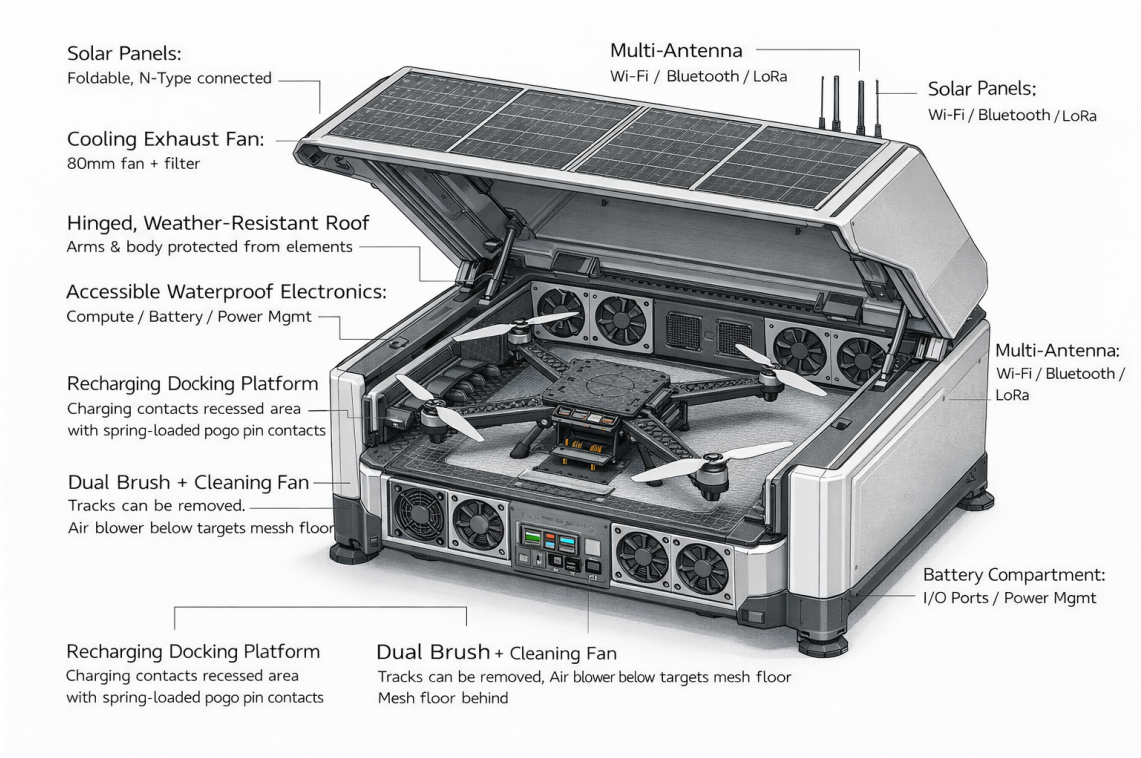

Figure 2. Buildable base station concept with docking, charging, cooling, and communication zones.

5. Mechanical layout notes

Drone frame target: medium 450-500 mm class. Keep the Raspberry Pi 5 on the top deck or upper center stack, camera in front, optical flow and rangefinder underneath, and pogo contacts centered under the body. For the base station, keep the charging pad centered with guide rails or funnel walls so the drone can settle naturally onto the contact zone.

Recommended connection and placement rules

- Cube Orange and Raspberry Pi 5 should be connected by a short, locked-down serial or USB cable.
- Optical flow sensor must point straight down and remain rigid.
- Rangefinder must also point straight down and be kept clear of prop wash.
- Camera should be front-facing or slightly downward if the mission is detection.
- Pogo pins should have a recessed docking bay and enough spring travel to tolerate 2-4 mm misalignment.
- Use a dedicated charge interlock so charging cannot start while the drone is armed.
- Keep the Pi in a lightweight aluminium heatsink case with active cooling.

6. 2-day build plan

Day	Main objective	What must be working by the end
Day 1	Flight stack + no-GPS stabilization	Cube Orange, Mission Planner, optical flow, rangefinder, arming, hover, and safe landing
Day 2	Detection + logging + transfer + docking logic	Camera feed, OpenCV detection, image save, rsync/scp transfer, and docking/charge interlock logic

7. Final must-buy / must-use list

Must-buy	Reason
Cube Orange	Flight control and no-GPS stability
Raspberry Pi 5	Camera, detection, logging, and transfer
Optical flow sensor	Indoor X-Y hold without GPS
Rangefinder	Altitude hold and landing
Camera module	Detection and image capture
Pogo pins + charger stage	Dock recharge path
24V 5A SMPS	Base station charging source
Wi-Fi network	Image/log transfer to base

This document intentionally avoids GPS, ROS, SLAM, and onboard LLMs so the team can finish a functioning system inside 2 days. The build path is to get the airframe flying, stabilize hover with optical flow plus rangefinder, then add detection and transfer.

8. Practical conclusion

If you follow the stack above, you get a buildable and defensible system: Cube Orange for flight, Pi 5 for detection, optical flow and rangefinder for no-GPS hold, and a pogo-pin base station with 24V 5A supply and 16.8V CC/CV charging stage. That is the shortest path to a real demo.